Predicting Severe Educational Delay in Rural India: A Computational Study Using Pooled IHDS Data

Introduction and Research Question -

Being the world's 5th largest and fastest growing economy, India has been striving towards economic growth and development for several years. Numerous policies have been devised and implemented over the past couple of decades specifically to this effect. One major point of focus has been the improvement in the level of education for rural children. Education is a major determinant of human capital formation and therefore is essential for sustained long-term economic growth.

In India, most rural children do enroll in primary school but due to various reasons their progression is irregular which in most cases makes dropping out more prevalent. As a result, the identification of specific policies which enable children to remain in the correct grade for age is important in the context of human capital formation and therefore the broader objective of economic growth and development.

The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) was passed in 2005 and is the policy under consideration. MGNREGA promises at least 100 days of employment for every rural household for a specific wage. MGNREGA has a two-pronged approach, first to provide unemployed members of rural households with work and second, the completion of work provided in turn, would socially and economically elevate these rural people themselves.

By providing income and employment definitively to rural households, MGNREGA may have a positive effect on the education of rural children and more specifically on the probability of them being in the right grade for age. This is because children will not have to supplement parent labor income any longer and will also need to contribute less to agricultural work and resource collection. On the other hand, if household employment opportunities rise, children may have to substitute into household work. As a result, the effect of MGNREGA on the education of children is ambiguous and needs to be empirically tested.

The objective of this project is not only to keep that policy setting and grade for age focus but to also extend the analysis beyond an average effect framework. The benchmark difference-in-differences section is used to estimate average programme-related change in schooling progression, whereas the machine-learning section will be used to identify which children face the highest risk of severe educational delay. This will allow the project to explore simultaneously whether average progress improved and whether serious educational vulnerability remains concentrated among specific groups.

Research Question: To what extent is village-level exposure to MGNREGA associated with changes in children's grade-for-age progression in rural India, and which child, household, and regional characteristics best predict severe educational delay, defined as being two or more grades behind the expected grade for age?

Data Description and Processing Steps -

The IHDS data jointly collected by the University of Maryland and the National Council of Applied Economic Research has been used for this analysis. The IHDS was conducted as a nationally representative survey for two waves (2004-05 and 2011-12) containing detailed information on household characteristics, village context, and children's schooling and education. As a result, IHDS is uniquely suited to undertake a before and after analysis in the context of MGNREGA.

In each wave I retain rural households from the household files and construct per capita monthly consumption expenditure and the highest adult completed years of schooling in the household. I use the 2011-12 household file to identify whether any household member reports income or work hours from MGNREGA. I then use this information at the village level by individually looking at each village, counting the number of households surveyed and how many of them reported MGNREGA participation.

A village is defined as treated if at least one household from it reported work or income from MGNREGA and control if no such work or income was reported by any household from that village. This treatment status is then attached to all children living in those villages in both waves. For observations from 2011-12 and 2004-05 a time dummy post equals 1 and 0 respectively. Since the 2004-05 wave refers to a period before MGNREGA was implemented, all villages are effectively untreated in the pre-period even if they are classified under treated based on their future participation.

Child level information is obtained from the individual files and in each wave, children are merged to their household records with the help of the household ID's. I restrict the sample by only considering rural children aged 6-14, keeping those with non-missing information on age, current grade of schooling, and key controls. Age is measured in completed years and in each wave, grade is recorded separately. Negative or special values in the grade variables are recorded as missing and then the expected grade variable and ontrack indicator are constructed. To enable the usage of same notation across waves, a generic grade variable is defined and the grade gap outcome is derived.

The final pooled samples consist of 28,661 rural children between the ages of 6 and 14 in 2004-05 and 25,439 in 2011-12 living in 1,417 rural villages across the Indian states. Their summary statistics reveal that both treatment and control villages are similar to each other in the context of age and gender composition. However, it can be observed from the summary statistics that villages which later adopt MGNREGA start out poorer and with a lower level of adult education than villages which do not adopt MGNREGA.

For the benchmark DiD section, the key schooling measures are the on-track indicator and the grade-gap outcome constructed from actual grade and expected grade. For the machine-learning section, I additionally construct a severe educational delay indicator from the same grade-for-age information already present in the pooled dataset. Severe educational delay is defined as being two or more grades behind the expected grade for age. I do this because this creates a sharper policy target than the simple on-track indicator by focusing specifically on children experiencing more serious schooling delays.

Methods and Why They Are Appropriate -

The discussion in the previous section suggests that MGNREGA may affect children's schooling in a way such that it is essentially an optimization problem wherein parents must choose how to allocate a child's time subject to budget and time constraints. MGNREGA shifts this optimization problem by raising income and potentially changing the opportunity cost of children's time.

The first method used in this project is a benchmark DiD model. Children's grade for age is the outcome of interest for the empirical DiD model. For each child from village v and survey year t, I observe their completed grade and age in years. I then construct the expected grade assuming entry into class 1 at age 6 as expected_grade = age – 5, and define the binary indicator ontrack = 1(grade ≥ expected_grade), which equals 1 if the child is at least in the expected grade for their age and 0 otherwise. Both late entry and repetition are captured in this variable. As a robustness check, I use a continuous grade gap instead of ontrack as an outcome, and define it as gradegap = grade – expected_grade, which measures by how many grades a child is ahead or behind the expected grade for age.

The DiD estimate refers to by how many percentage points the probability of a child being on track changes over time in MGNREGA villages compared to other similar non-MGNREGA villages using the specified controls and state fixed effects. One major assumption of this model is the parallel trends assumption. In this context, the parallel trends assumption refers to the fact that in the absence of MGNREGA, treated and control villages would have seen similar changes in grade-for-age conditioned on the included controls.

This benchmark DiD section is appropriate because the pooled dataset has an explicit pre-period and post-period structure and treatment is defined at the village level. As a result, it provides a transparent average-effect estimate of programme-related change in schooling progression.

The second method used in this project is supervised machine learning for prediction. The prediction task is to classify whether a child is severely educationally delayed. This method is appropriate because as discussed before, severe educational delay is a sharper policy outcome than the simple on-track indicator and is more directly connected to severe educational vulnerability.

Predictor variables include age, age squared, gender, household consumption, adult education, survey period, village treatment status, and state identifiers. Logistic regression is used as a benchmark classifier, while tree-based methods such as random forest are used because they can handle high-dimensional non-linear data, are robust to multicollinearity, and provide interpretable measures of feature importance. Model performance is evaluated using metrics such as ROC-AUC and F1, and feature-importance measures are used only as indicators of predictive relevance rather than causality. This makes the machine-learning section appropriate for this project because it answers a different but complementary question to the benchmark DiD. Specifically, the DiD section estimates whether average grade progression improved more in treated villages over time, while the machine-learning section identifies which children remain at greatest risk of serious educational lag.

Results -

Table 1- Balance Table for 2004-2005

Variable	Control Villages (treat_v = 0)	Treated Villages (treat_v = 1)
Age (years)	10.05	9.95
Girl (Dummy)	0.476	0.480
ontrack	0.305	0.228
COPC	681.84	585.01
Highest Adult Education (years)	6.37	5.43
Observations	8,981	19,680

Table 2- Mean on-track rates by treatment status and period

Group	2004-05 (post=0)	2011-12 (post=1)
Control Villages (treat_v = 0)	0.305	0.340
Treated Villages (treat_v = 1)	0.228	0.310

Table 1 is suggestive of the fact that treated villages start off as poorer and have lower levels of adult education, while child age and gender distributions are very similar across the treatment and control groups. This confirms the need to control for household consumption expenditure and adult education while estimating the main DiD effect.

From Table 2 it is observed that children in MGNREGA villages experience a higher improvement in grade for age compared to those in non-MGNREGA villages. In 2004–05 only about 22 per cent of children in MGNREGA villages were on track for age, whereas, by 2011–12 about 31 per cent of them were on track for age. Comparatively, children in non-MGNREGA villages move from about 30 per cent to 34 per cent. Therefore, though both groups improve over time, MGNREGA villages improve more in absolute terms thereby implying a raw DiD of around 4.7 percentage points.

Benchmark DiD estimates and Grade-Gap Robustness Estimates -

Table 3 -

Table 3 describes the main DiD regressions, where the outcome is the indicator for being on track for age, together with the continuous grade-gap robustness check. Age, age squared, per-capita consumption (COPC), highest adult education and state fixed effects are included in both specifications, and the standard errors are clustered at the village level.

Model	**MGNREGA village (treat_v)**	**post**	**treat × post**	**girl**	**Observations**	**R2**
On track	-0.0367	0.0201	0.0628	0.0121	54058	0.099
Grade gap	-0.119	0.196	0.251	-0.0493	54058	0.251

The key parameter treat × post is highly significant and is around 0.063. Therefore, the probability of being on track for age increased by about 6–7 percentage points more in MGNREGA villages. In the grade-gap specification, the coefficient on treat × post is 0.251 which implies that children in MGNREGA villages end up around a quarter of a class closer to where they are supposed to be in comparison to similar children in non-MGNREGA villages.

ML Section -

Table 4 – Model Comparison for Severe Educational Delay Prediction

model	**best_cv_avg_precision**	**test_accuracy**	**test_roc_auc**	**test_avg_precision**	**test_precision**	**test_recall**	**test_f1**
Random Forest	0.720	0.733	0.807327172	0.728	0.698	0.582	0.634
Logistic Regression	0.696	0.7196	0.784	0.695	0.667	0.591	0.627

The machine-learning section is designed to focus on severe educational delay, Among the classifiers considered, Random Forest performs best, with a test ROC-AUC of 0.807, Average Precision of 0.728, accuracy of 0.733 and F1 score of 0.635. These values indicate that severe educational delay can be predicted with reasonable precision and that this sharper target is more informative than a broader on-track classification.

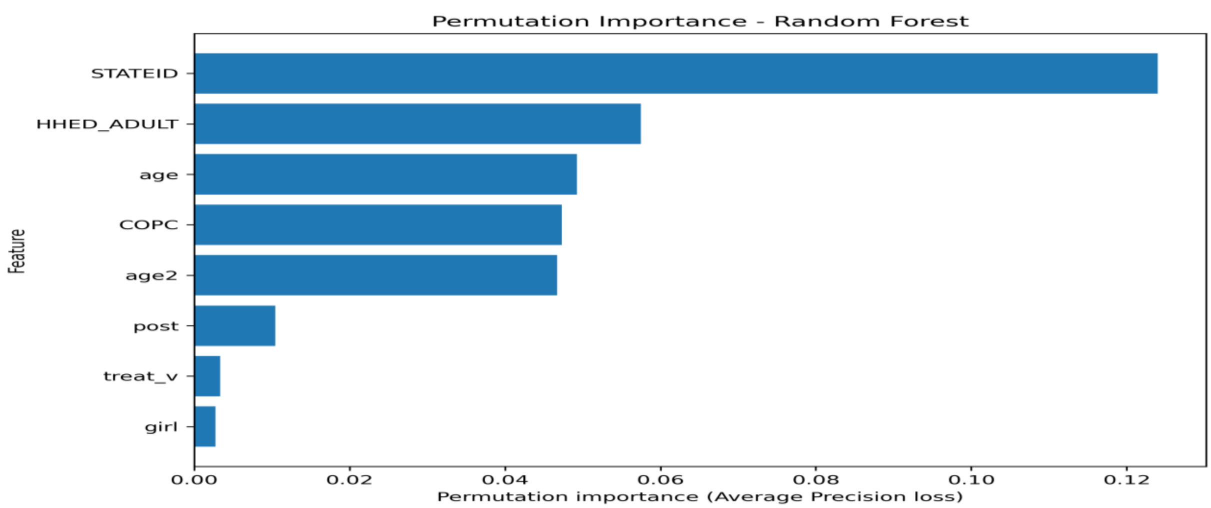

Figure 1. Permutation Importance of Predictors of Severe Educational Delay

Table 5 – Permutation Importance Scores for severe educational delay

feature	importance_mean	importance_std
STATEID	0.12402251	0.004236338
HHED_ADULT	0.057493617	0.003136037
age	0.04925072	0.001524412
COPC	0.047327295	0.001451926
age2	0.04668962	0.001517107
post	0.010469244	0.000752192
treat_v	0.003384768	0.000960236
girl	0.002728923	0.000535713

Permutation importance shows that the strongest predictors of severe educational delay are STATEID, HHED_ADULT, age, COPC and age2, while post, treat_v and girl play a smaller role in prediction. This suggests that severe delay is driven mainly by structural disadvantage and regional heterogeneity rather than by one programme variable alone.

Table 6 – Severe Educational Delay by subgroup

groupingvariable	Group value	n	Actual severe rate	Mean predicted severe prob	Median predicted severe prob	Mean years behind	Median years behind
gender_group	Boys	8476	0.399	0.395	0.382	1.501	1
gender_group	Girls	7742	0.398	0.400	0.390	1.547	1
age_group	Age 11-14	7270	0.522	0.515	0.500	2.121	2
age_group	Age 6-10	8948	0.298	0.302	0.294	1.037	1
consumption_group	Higher consumption	8108	0.345	0.350	0.333	1.297	1
consumption_group	Lower consumption	8110	0.452	0.445	0.458	1.749	1
education_group	Higher adult education	7360	0.305	0.315	0.305	1.083	1
education_group	Lower adult education	8858	0.476	0.4667	0.472	1.888	1
village_group	MGNREGA village	11215	0.429	0.423	0.427	1.629	1
village_group	Non-MGNREGA village	5003	0.329	0.341	0.325	1.284	1

Severe-delay risk is higher among older children, lower-consumption households, lower-adult-education households and children in MGNREGA villages.

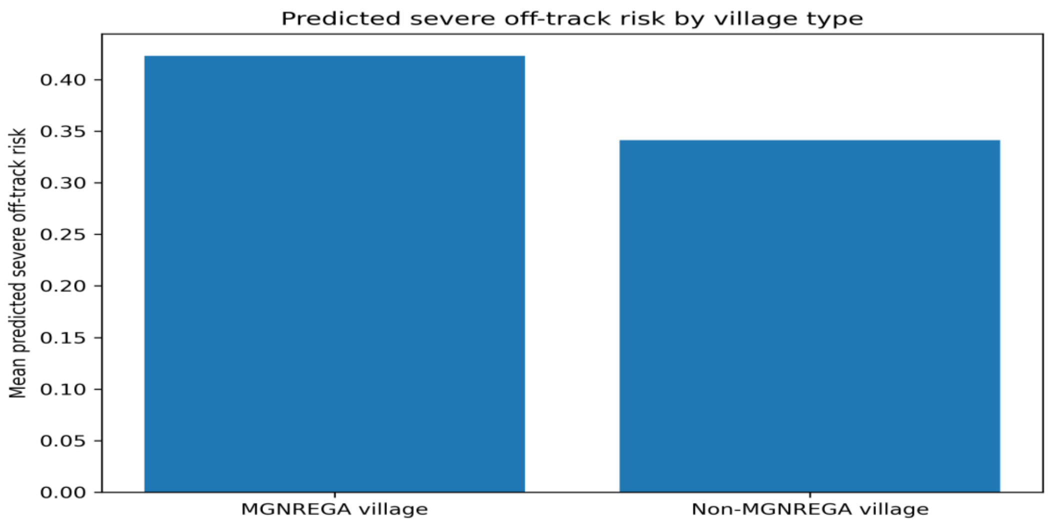

Figure 2. Predicted Severe Educational Delay Risk by Village type

Table 7. High-Risk Profile of Children with the Highest Predicted Severe Delay

metric	value
n_highest_risk	3244
share_girls	0.502774358
mean_age	11.76263905
median_age	12
mean_COPC	3745.125555
median_COPC	478
mean_HHED_ADULT	2.832305795
median_HHED_ADULT	0
share_MGNREGA_village	0.84802711
actual_severe_rate	0.778360049
mean_years_behind	3.225030899
median_years_behind	3
mean_predicted_severe_prob	0.728563732
median_predicted_severe_prob	0.717035499

The highest-risk group is concentrated in disadvantaged settings. Around 84.8 percent are in MGNREGA villages, median household consumption is 478, median adult education is 0, median years behind is 3, and the actual severe-delay rate is about 77.8 per cent. Taken together, the results suggest that while average grade progression improves more in MGNREGA villages over time, serious educational lag remains concentrated among structurally vulnerable children.

Interpretation, Limitations and Policy Implications -

The results indicate a coherent two-part story. First, the benchmark estimate shows that children in MGNREGA villages experienced a greater improvement over time compared to those in non-MGNREGA villages. The implication of the treat × post estimate is consistent with the idea that public employment programmes may diminish household constraints and improve children's schooling progression as a result, even when it may not have been the direct target.

Second, the Machine Learning results demonstrate the concentrated existence of severe educational vulnerabilities in structurally disadvantaged children. Severe educational delay is predicted majorly by state, adult education, age, and household consumption, whereas village treatment status plays a much smaller role in prediction. The highest-risk children are disproportionately older, concentrated in low-education households and heavily represented by MGNREGA villages. This is by no means a contradiction of the benchmark DiD result. It is instead suggestive of the fact that treated villages can improve more over time on average while containing a larger concentration of vulnerable children because they begin from a more disadvantaged baseline. The descriptive evidence is also consistent with this pattern.

These findings should, however, be interpreted with a certain amount of caution. The DiD design is heavily reliant on the parallel trends assumption, which cannot be directly tested in a richer environment or dynamic method using only the two survey waves. Unobserved differences between treated and control villages may also remain despite controlling household characteristics and state fixed effects. It must also be noted that the machine-learning section is predictive and not causal. It is designed such that it identifies which observed characteristics are most useful for predicting severe educational delay, but it does not establish that those characteristics themselves may cause educational failure. To reiterate, the feature-importance findings indicate predictive relevance rather than causal effect.

There are two major policy aspects that are worth discussing, At the average level, the DiD results tend to indicate that MGNREGA positively affect/lead to, children's educational progression in India. The machine learning results, however, show how average improvement does not mitigate concentrated educational risk. Severe delay remains common specifically in the context of older children and seriously disadvantaged households. This implies that even though broad rural employment may improve schooling progression, additional targeted interventions are essential for most educationally vulnerable children.

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